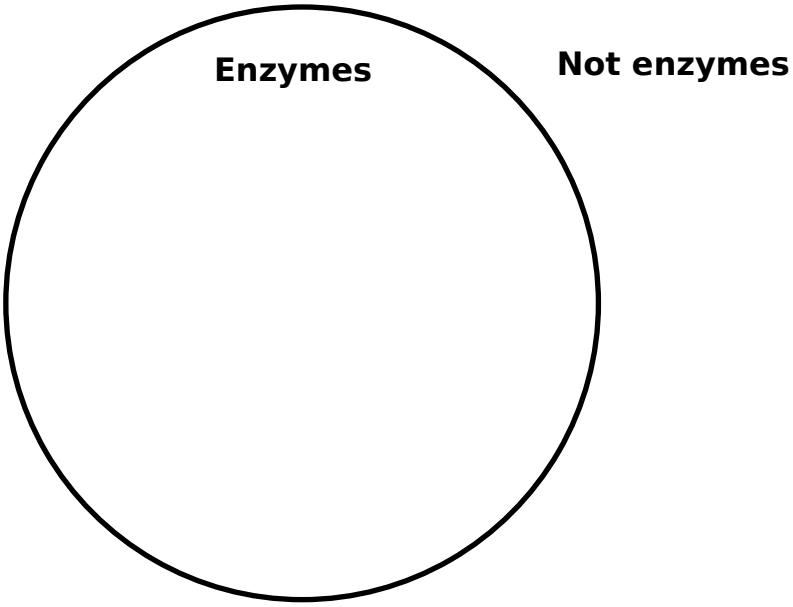
Starter task

Put the chemicals into the correct space on the Venn diagram.

amylase
lipase
glucose
pyruvate
protease
pepsin
reductase
maltose
glycerol
ligase
lactose

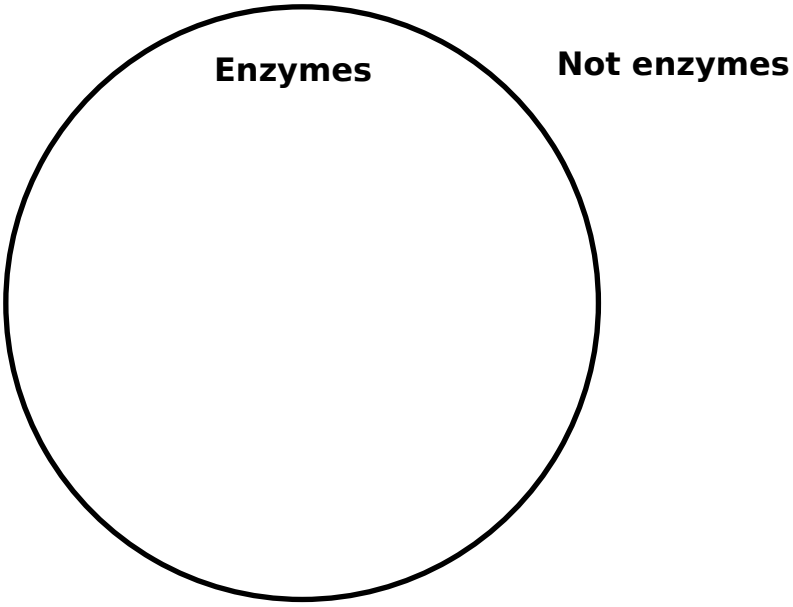


1. What is the general rule about the suffix used for enzymes?
2. Are there any exceptions to the rule?

Starter task

Put the chemicals into the correct space on the Venn diagram.

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1. What is the general rule about the suffix used for enzymes?
2. Are there any exceptions to the rule?

Information sheet

Task 1

Quickly skim read the text, to get the ‘gist’ of each paragraph. Write in the box above each paragraph what you think it is about in 5 words or less!

A

Your liver is important for breaking down any potentially dangerous substances you consume.

Within your liver, as within every tissue in the body, many chemical reactions occur.

Often these reactions require "help" to happen at a faster speed, and this can be supplied by enzymes — types of proteins.

Enzymes speed up the rate of a chemical reaction by acting as a catalyst.

A catalyst is a substance that increases the rate of a chemical reaction without itself undergoing any permanent chemical change.

B

All enzymes are made of proteins and have a specific shaped site where the chemical reaction can take place.

This is known as the enzyme’s active site.

Each enzyme has a different shape active site which explains why they only interact with specific reactants (which are known as substrates).

We call the ‘fit’ between the active site and the substrate complementary.

C

Enzymes are very specific and need to be in certain environments or conditions to work well —or at all.

If the temperature is too low, then the enzymes and substrates do not have very much kinetic energy, so they don’t collide very often.

This means that the rate of reaction is slow.

Increasing the temperature will increase the kinetic energy so will increase the rate of reaction.

However, enzymes can be damaged (denatured), if exposed to too much heat.

A denatured enzyme may no longer work to catalyze a chemical reaction because the heat will alter the shape of the active site.

All enzymes have ‘optimum’ conditions – this means they each have a temperature and pH where they work at a maximum rate.

D

Catalase is an example of an enzyme in the liver that breaks down harmful hydrogen peroxide into oxygen and water.

When this reaction occurs, oxygen gas bubbles escape and create foam.

Task 2

Look at the information sheet again. Use scanning techniques to highlight the answers to the questions below.

1. What is an enzyme?
2. What is a catalyst?
3. Describe the structure of an enzyme.
4. What is a substrate?
5. Give an example of a condition which might prevent an enzyme working.
6. When might an enzyme become denatured?
7. Give one way the rate of enzyme controlled reactions be increased?
8. Name an enzyme found in the liver.
9. Name the substrate that the enzyme named in question 8 breaks down.

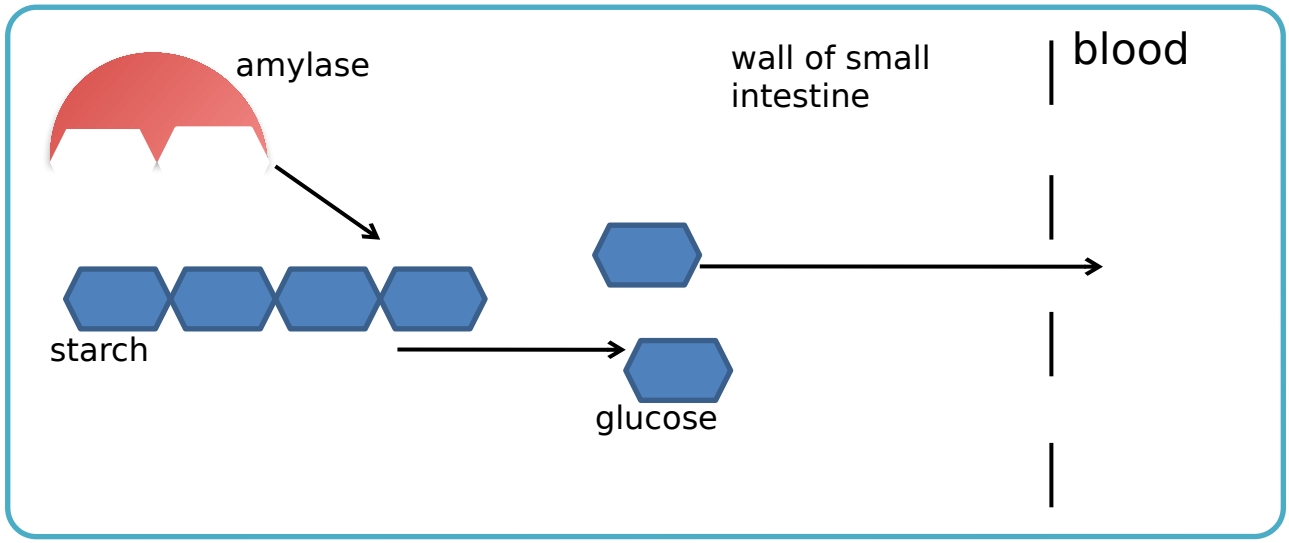
Task 2

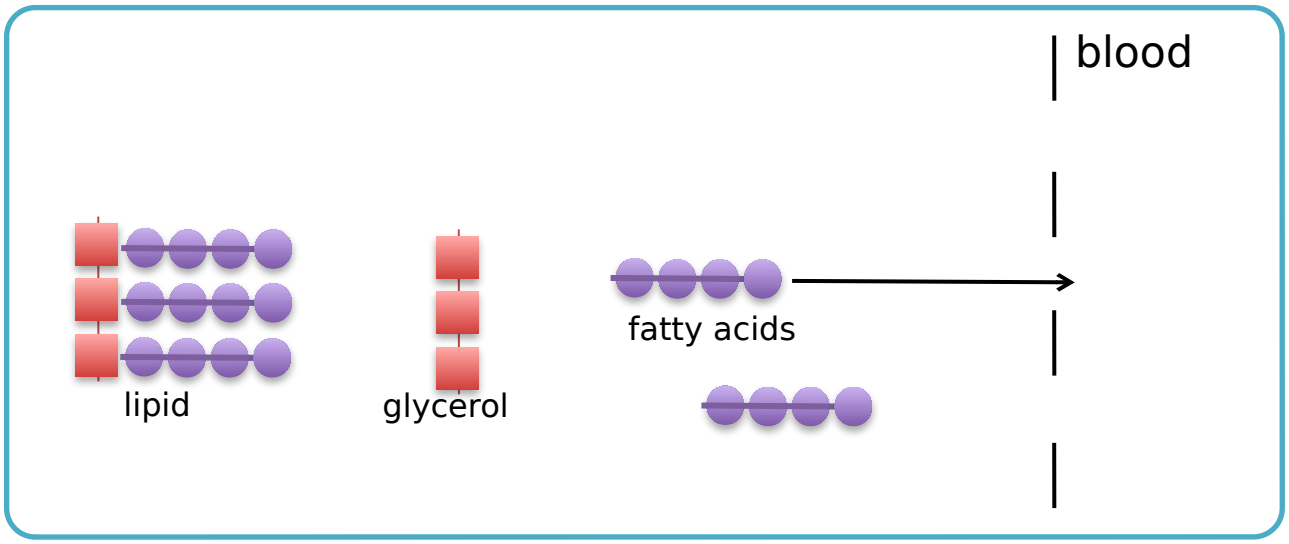
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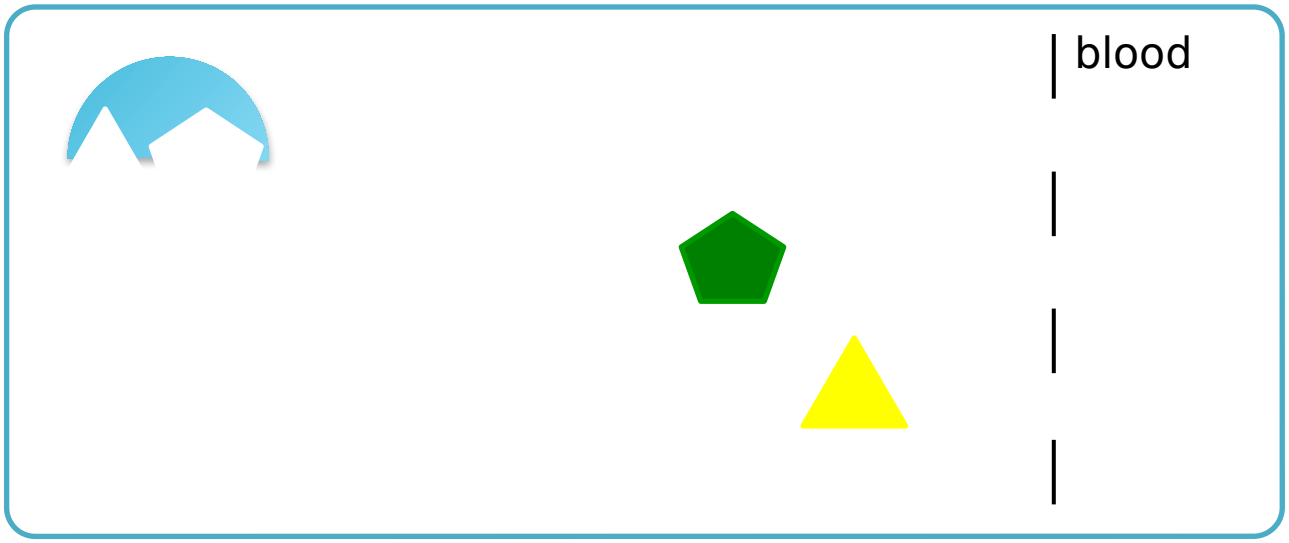
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Task 3 – how do digestive enzymes work?

Complete the series of diagrams below to show how 3 different digestive enzymes work. You will need to label and annotate the diagrams.







Check your knowledge

Task 4

1. Define the term catalyst.

.....

.....

2. Define the term enzyme.

.....

.....

3. What does each enzyme do? Complete the sentences about specific enzymes.

a. ..... breaks down lipids (fats) to form ..... and glycerol.

b. Amylase breaks down ..... to form .....

c. ..... breaks down proteins to form ....

4. A student eats steak for dinner. Steak contains protein and fat. Explain why amylase does not work to break down either fat or protein.

Keywords to include:

active site	fit	substrate	protein	starch

.....

.....

.....

.....

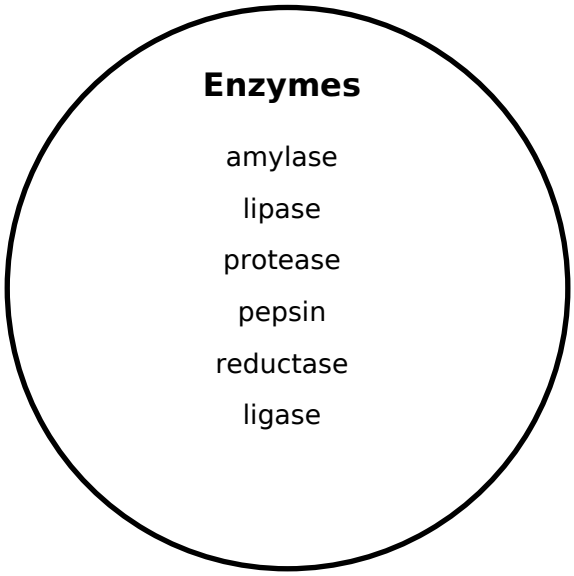
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5. Label the diagram.



Teaching notes and answers

Starter task



Not enzymes

glucose
pyruvate
maltose
glycerol
lactose

1. They have the suffix -ase
2. Pepsin

Task 2

1. An enzyme speeds up the rate of a specific reaction, without being used up.
2. A catalyst is a substance that increases the rate of a chemical reaction without itself undergoing any permanent chemical change.
3. All enzymes are proteins and have a specific shaped active site.
4. The reactant that is broken down by the enzyme.
5. High temperature
6. When exposed to a high temperature
7. Increasing temperature up to the optimum for that enzyme.
8. Catalase
9. Hydrogen peroxide

Task 3

Students should include labels for the enzyme, active site, substrates and products. They should annotate to describe the enzyme-substrate complex formed during digestion. They should clearly show that the shape of the active site is specific to the shape of the substrate. The first diagram shows the digestion of starch, the second, fats and the third, proteins.

To differentiate the task, give students who need more support just the first diagram to complete. For students who need more challenge, ask them to draw their own diagram to illustrate the breakdown of proteins and/or fats and/or starch without the sheet.

Task 4

1. A catalyst is a substance that increases the rate of a chemical reaction without itself undergoing any permanent chemical change.
2. An enzyme speeds up the rate of a specific reaction, without being used up.
3. What does each enzyme do? Complete the sentences about specific enzymes.

   a. *Lipase* breaks down lipids (fats) to form *fatty acids* and glycerol.
   b. Amylase breaks down *starch* to form *maltose or simple sugars*.
   c. *Protease* breaks down proteins to form *amino acids*.

6. Enzymes only work on a specific substrate. This is because they have an active site which only fits one type of substrate. Amylase is a carbohydrase – it breaks down starch, but not proteins or fats. Lipase and protease are the enzymes which would need to be present to break down the fats and proteins.
7. 